Quantum Computing

Chapter 03: Quantum Algorithm and State Preparation

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Who? Me?

- Nickname: Arm (P'/N' Arm, etc.)
- Born: Aug 1981
- Work
  - Researcher at NECTEC 2005-2024
  - Lecturer at SIIT, Thammasat University 2025-now
- Education
  - B.Eng & M.Eng in Computer Engineering, Kasetsart University, Thailand
  - Obtained Ministry of Science and Technology Scholarship of Thailand in early 2008
  - Did a PhD in Informatics (AI & Computational Linguistics) at University of Edinburgh, UK from 2008 to 2013

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Introduction

Quantum Algorithm

- Quantum algorithm is a circuit of quantum logic gates for manipulating qubit-based information to achieve desired computation
    1. Preparation: convert input into quantum states, superpositions, and entanglement
    2. Computation: via state manipulation, conditions, loops, reflections, and phase estimation
    3. Measurement: measure the probability distribution for each quantum state

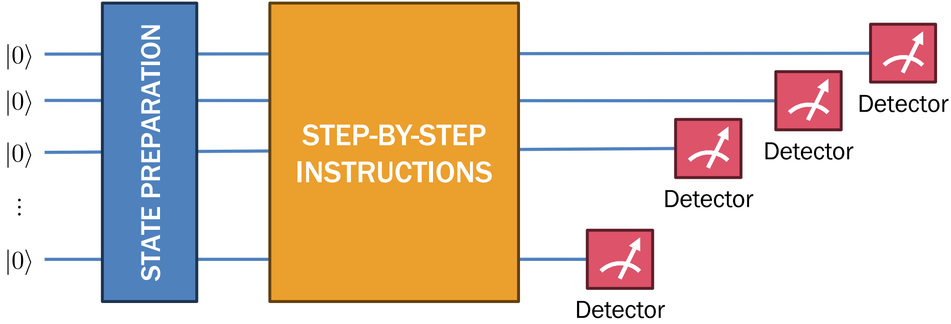


Figure 1: Overview of quantum algorithm

PennyLane

- Popular library for quantum computing in Python developed by Xanadu
- Features:
  - Easy to use with low learning curve
  - Support for CPU-based and GPU-based quantum simulation
  - Plugins for actual QPUs e.g. IBM Qiskit, Amazon Braket, and Xanadu
  - Support for automatic JIT compilation of quantum algorithms in the entire quantum-classical workflow
- The URL is `https://pennylane.ai/`

- Basic installation

```
pip install pennylane
```

- On CUDA-enabled machine

```
pip install custatevec_cu12
pip install pennylane-lightning-gpu
```

- On CUDA Tensor Networks

```
pip install cutensornet-cu12
pip install pennylane-lightning-tensor
```

- Docker

```
docker pull pennylaneai/pennylane:latest-
    lightning-qubit
```

Example Code

```python
import pennylane as qml                # Quantum computing
from pennylane import numpy as np      # Classical computing

# Quantum device with two wires (i.e. variables)
dev = qml.device('default.qubit', wires=[0, 1])

# This is our simple quantum algorithm
@qml.qnode(dev)
def circuit(theta: float):

    # State preparation
    qml.H(wires=[0])                # Hadamard gate on qubit-0

    # Computation
    qml.RY(theta, wires=[0])        # Rotate qubit-0 on Axis-Y by theta
    qml.CNOT(wires=[0,1])           # Controlled NOT on qubit-1 with control qubit-0

    # Measurement
    return qml.state()

# Let's run our quantum algorithm
print(circuit(np.pi / 6))
```

Code Skeleton

Header

```
import pennylane as qml
from pennylane import numpy as np
```

Quantum device

```
dev = qml.device('default.qubit', wires=...)
```

Several quantum devices are also available.

- `default.qubit`: Simple simulator
- `default.mixed`: Mixed-state simulator
- `lightning.qubit`: C++-based simulator
- `default.tensor`: Tensor Networks

Quantum algorithm = quantum node

- Preparation = variable initialization
- Computation = sequence of logic gates
- Measurement = collapsing quantum states to a bitstring

```
@qml.qnode(dev)
def circuit(args):
    # State preparation
    ......

    # Computation
    ......

    # Measurement
    return ......
```

Clifford Set/1

Clifford Set of Quantum Logic Gates

- A set of basic quantum logic gates
  - Fault-tolerant: Hardware implementation of these gates can significantly tolerate noise and quantum decoherence
  - Self-inversible: Applying a quantum operator twice on a state results in the original one
  - Efficiently simulatable: They can be emulated in polynomial time by probabilistic classical computers
- The main purpose of the Clifford set is state preparation
- A quantum algorithm consists of only Clifford gates is sometimes called stabilizer circuit due to its stability
- In this chapter, we will divide them into three levels

Level 1	Level 2	Level 3
NOT Hadamard Phase shift Controlled NOT	Pauli X Pauli Y Pauli Z	Swap Controlled Z

Table 1: Clifford set of quantum logic gates

- NOT gate converts $|0\rangle$ to $|1\rangle$ and vice versa, similar to logical NOT

$$\begin{aligned} \mathtt{X} &= |\overline{x}\rangle\langle x| \\ &= |1\rangle\langle 0| + |0\rangle\langle 1| \\ &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{aligned}$$

- $\mathtt{X}$ is self-inversible

$$\mathtt{X}\mathtt{X}|x\rangle = |x\rangle$$

- $\mathtt{X}$ is often used for preparing $|1\rangle$
- `qml.X(wires=...)`

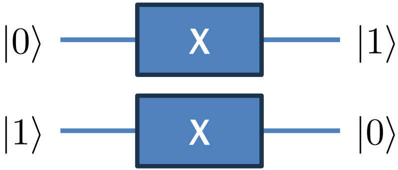


Figure 2: NOT gate

Level 1: Hadamard Gate

- Hadamard gate converts pure states to superpositions and vice versa

$$
\begin{aligned}
\mathbf{H} &= |+\rangle\langle 0| + |-\rangle\langle 1| \\
&= \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}
\end{aligned}
$$

- $\mathbf{H}$ is self-inversible

$$\mathbf{H}\mathbf{H}|x\rangle = |x\rangle$$

- Hadamard gate is often used for preparing a superposition
- `qml.H(wires=…)`

- Let's define the superposition bases

$$
\begin{aligned}
|+\rangle &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \\
|-\rangle &= \frac{|0\rangle - |1\rangle}{\sqrt{2}}
\end{aligned}
$$

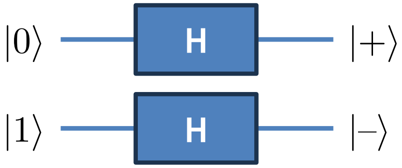


Figure 3: Hadamard gate

Level 1: Phase Shift Gate

- Phase shift gate converts real numbers to imaginary numbers and vice versa

$$\begin{aligned} \mathtt{S} &= |0\rangle\langle 0| + \left(i\,|1\rangle\right)\langle 1| \\ &= \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \end{aligned}$$

- $\mathtt{SS}$ is equivalent to sign flipping

$$\begin{aligned} \mathtt{SS}\,|0\rangle &= |0\rangle \\ \mathtt{SS}\,|1\rangle &= -\,|1\rangle \end{aligned}$$

- `qml.S(wires=…)`

- Let's define the circular bases

$$\begin{aligned} |{+i}\rangle &= \frac{|0\rangle + i\,|1\rangle}{\sqrt{2}} \\ |{-i}\rangle &= \frac{|0\rangle - i\,|1\rangle}{\sqrt{2}} \end{aligned}$$

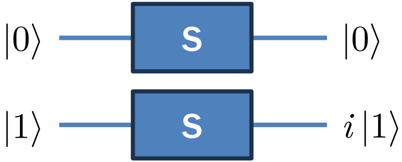


Figure 4: Phase shift gate

Level 1: Controlled NOT

- Controlled NOT is a condition tester, where the second qubit is bit-flipped if the first qubit is $|1\rangle$

$$\begin{aligned}\mathrm{CNOT} &= |0,x\rangle\langle 0,x| + |1,\overline{x}\rangle\langle 1,x| \\ &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}\end{aligned}$$

- CNOT is also self-inversible

$$\mathrm{CNOT}\,\mathrm{CNOT}\,|x,y\rangle = |x,y\rangle$$

- `qml.CNOT(wires=…)`

- CNOT is equivalent to logical XOR $\oplus$

$$\mathrm{CNOT} = |x,(x\oplus y)\rangle\langle x,y|$$

- CNOT combined with $\mathbf{H}$ creates various forms of entanglement

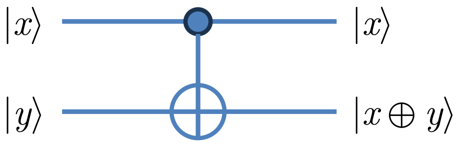


Figure 5: Controlled NOT gate

Summary of Level 1

Gates	Descriptions	Symbols = Formulae
NOT gate	Bit-flip the input qubit, similar to the logical NOT.	$\mathbf{X} = \lvert\overline{x}\rangle\langle x\rvert$
Hadamard gate	Convert pure $\leftrightarrow$ superposition on the input qubit.	$\mathbf{H} = \lvert+\rangle\langle 0\rvert + \lvert-\rangle\langle 1\rvert$
Phase shift gate	Convert real $\leftrightarrow$ imaginary if the input qubit is 1.	$\mathbf{S} = \lvert 0\rangle\langle 0\rvert + (i\lvert 1\rangle)\langle 1\rvert$
CNOT gate	Bit-flip the second qubit if the first one is 1. Create an entangled state when used with the Hadamard gate.	$\mathbf{CNOT} = \lvert 0,x\rangle\langle 0,x\rvert + \lvert 1,\overline{x}\rangle\langle 1,x\rvert$

Table 2: Basic Clifford gates

Special Unitary Group

- Special unitary group: $\mathbf{SU}(n)$ is the set of $n \times n$ complex matrices that are unitary and special
- Formal defintion:

$$\begin{aligned} \mathbf{SU}(n) \quad &= \quad \{\mathbf{U} \in \mathbb{C}^{n \times n} | \mathbf{U}\mathbf{U}^* = \mathbf{I} \\ &\qquad \text{and } \det(\mathbf{U}) = 1\} \end{aligned}$$

- Why mentioning this?
  - Unitary $\mathbf{U}\mathbf{U}^* = \mathbf{I}$: All of these matrices are quantum operators
  - Special $\det(\mathbf{U}) = 1$: All of these matrices do not change the global phase

Quantum Gates	Groups
NOT gate	$\mathbf{SU}(2)$
Hadamard gate	$\mathbf{SU}(2)$
Phase shift gate	$\mathbf{SU}(2)$
CNOT gate	$\mathbf{SU}(4)$

Table 3: Clifford set and their SU groups

Quantum Algorithm: Example 1

- Step 0: The initial state is always $|0\rangle^{\otimes N}$.

$$\left|s^{(0)}\right\rangle = |0\rangle \otimes |0\rangle$$

- Step 1: Apply the Hadamard gate on the first qubit.

$$\begin{aligned}\left|s^{(1)}\right\rangle &= (\mathbf{H}\,|0\rangle) \otimes |0\rangle \\ &= |+\rangle \otimes |0\rangle\end{aligned}$$

- Step 2: Applying the NOT gate on the second qubit.

$$\begin{aligned}\left|s^{(2)}\right\rangle &= |+\rangle \otimes (\mathbf{X}\,|0\rangle) \\ &= |+\rangle \otimes |1\rangle\end{aligned}$$

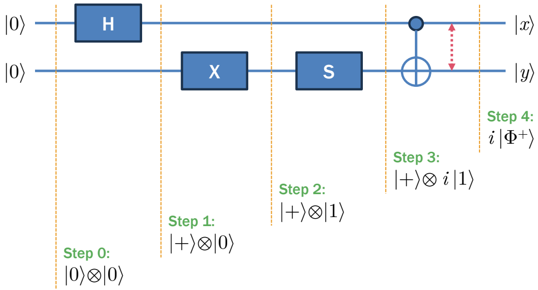


Figure 6: Just a quantum algorithm

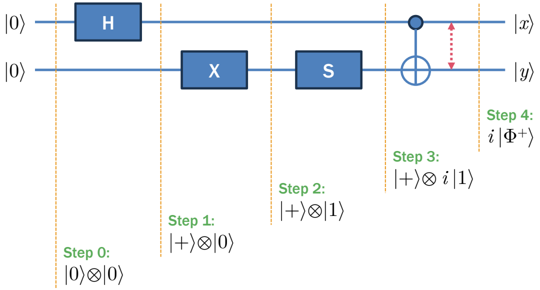
Quantum Algorithm: Example 1

- Step 3: Applying the phase shift gate on the second qubit $|1\rangle$ yields $i\,|1\rangle$.

$$\left|s^{(3)}\right\rangle = |+\rangle \otimes (\mathbf{S}\,|1\rangle) = |+\rangle \otimes i\,|1\rangle$$

- Step 4: To apply CNOT, we realize the tensor product of its inputs.

$$|+\rangle \otimes i\,|1\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes i\,|1\rangle = \frac{i}{\sqrt{2}}\left(|01\rangle + |11\rangle\right)$$

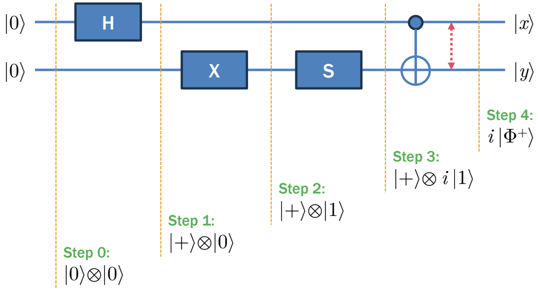


Quantum Algorithm: Example 1

- Step 4 (cont'd): Applying CNOT on $\frac{i}{\sqrt{2}}(|01\rangle + |11\rangle)$ yields

$$
\begin{aligned}
\left|s^{(4)}\right\rangle &= \text{CNOT}\left(\frac{i}{\sqrt{2}}(|01\rangle + |11\rangle)\right) \\
&= \frac{i}{\sqrt{2}}(|01\rangle + |10\rangle) \\
&= i\left|\Phi^{+}\right\rangle
\end{aligned}
$$

Note that the output now becomes an entangled state. The entanglement is denoted with a bidirectional arrow between the two qubits.



Example 1 in PennyLane

```
import pennylane as qml
from pennylane import numpy as np

dev = qml.device('default.qubit', wires=[0, 1])

@qml.qnode(dev)
def circuit():
    # State preparation
    qml.H(wires=[0])
    qml.X(wires=[1])
    qml.S(wires=[1])
    qml.CNOT(wires=[0, 1])
    # Measurement
    return qml.state()

print(circuit())
```

Result: $i\,|\Phi^+\rangle = \frac{i}{\sqrt{2}}\left(|01\rangle + |10\rangle\right) = \begin{bmatrix} 0 & \frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \end{bmatrix}^\top$

```
[0.+0.j   0.+0.707j    0.+0.707j    0.+0.j]
```

Hadamard Gate and For-Loop

- Hadamard gate can be used as a for-loop

$$
\begin{aligned}
\left|s^{(0)}\right\rangle &= |000\rangle \\
\left|s^{(1)}\right\rangle &= (\mathbf{H}\,|0\rangle) \otimes |00\rangle \\
&= |+\rangle \otimes |00\rangle \\
\left|s^{(2)}\right\rangle &= |+\rangle \otimes (\mathbf{H}\,|0\rangle) \otimes |0\rangle \\
&= |+\rangle \otimes |+\rangle \otimes |0\rangle \\
\left|s^{(3)}\right\rangle &= |+\rangle \otimes |+\rangle \otimes (\mathbf{H}\,|0\rangle) \\
&= |+\rangle \otimes |+\rangle \otimes |+\rangle \\
\left|s^{(4)}\right\rangle &= \mathbf{U}\left(|+\rangle \otimes |+\rangle \otimes |+\rangle\right) \\
&= \sum_{k=0}^{7} \mathbf{U}\,|\mathbb{B}_3(k)\rangle
\end{aligned}
$$

- $\mathbf{H}^{\otimes N}|0\rangle^{\otimes N}$ is equivalent to 2^N iterations in classical computing

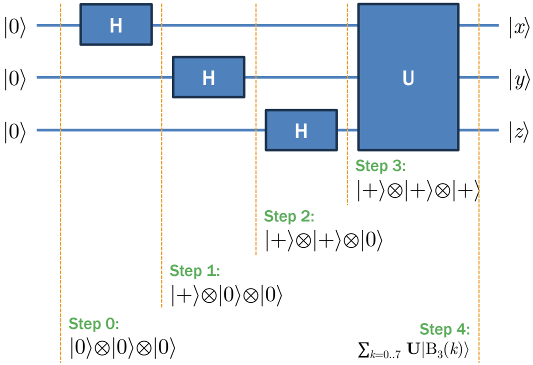


Figure 7: Quantum paralellism via superposition

Exercise 3.1: State Preparation

Design quantum algorithms that prepare the following states. Develop PennyLane code to verify each of them.

1. $|+\rangle$
2. $|01\rangle$
3. $|100\rangle$
4. $i\,|1\rangle$
5. $-|1\rangle$
6. $|-\rangle \otimes |+\rangle \otimes |-\rangle$
7. $|\Psi^+\rangle$
8. $|\Psi^-\rangle$
9. $|1010\rangle$
10. $|10\rangle \otimes |+\rangle \otimes |01\rangle$
11. $|\Phi^+\rangle$
12. $|\Phi^-\rangle$
13. $-|0\rangle$
14. $|01\rangle \otimes |\Psi^+\rangle$
15. $|\Psi^+\rangle \otimes |\Phi^-\rangle$
16. $(-|0\rangle) \otimes |1\rangle$
17. $|00\rangle \otimes |+\rangle \otimes |-\rangle$
18. $-|+\rangle$
19. $-|-\rangle$
20. $|0\rangle \otimes (-i\,|1\rangle)$
21. $|+\rangle \otimes |\Psi^+\rangle$
22. $(-i\,|+\rangle) \otimes |-\rangle$

Operator Augmentation

- If we have a quantum operator

$$\mathbf{Q} = \sum_{k=1}^{K} |\phi_k\rangle\langle\psi_k|$$

we can augment $\mathbf{Q}$ with trivial qubits by

$$\mathbf{I}^{\otimes M} \otimes \mathbf{Q} = \sum_{k=1}^{K} |x_1, \ldots, x_M, \phi_k\rangle\langle x_1, \ldots, x_M, \psi_k|$$

$$\mathbf{Q} \otimes \mathbf{I}^{\otimes M} = \sum_{k=1}^{K} |\phi_k, x_1, \ldots, x_M\rangle\langle\psi_k, x_1, \ldots, x_M|$$

- This is useful for accommodating all entangled qubits, some of which not involved with the operator

- Examples:

$$\begin{aligned}
\mathbf{I} \otimes \mathbf{X} &= |x, \overline{y}\rangle\langle x, y| \\
\mathbf{X} \otimes \mathbf{I} &= |\overline{x}, y\rangle\langle x, y|
\end{aligned}$$

$$\begin{aligned}
\mathbf{I} \otimes \mathbf{CNOT} &= |x, y, y \oplus z\rangle\langle x, y, z| \\
\mathbf{CNOT} \otimes \mathbf{I} &= |x, x \oplus y, z\rangle\langle x, y, z|
\end{aligned}$$

$$\begin{aligned}
\mathbf{I}^{\otimes 2} \otimes \mathbf{H} &= |x, y, +\rangle\langle x, y, 0| \\
&\quad + |x, y, -\rangle\langle x, y, 1| \\
\mathbf{I} \otimes \mathbf{H} \otimes \mathbf{I} &= |x, +, y\rangle\langle x, 0, y| \\
&\quad + |x, -, y\rangle\langle x, 1, y|
\end{aligned}$$

Operator Augmentation and Tensor Product

- Important property of tensor product

$$\left(\bigotimes_{k=1}^{N} \mathrm{A}_k\right)\left(\bigotimes_{k=1}^{N} \mathrm{B}_k\right) = \bigotimes_{k=1}^{N} \mathrm{A}_k \mathrm{B}_k$$

- Examples:

$$\begin{aligned}
(\mathrm{X} \otimes \mathrm{Y})(\mathrm{I} \otimes \mathrm{Z}) &= \mathrm{XI} \otimes \mathrm{YZ} \\
(\mathrm{H} \otimes \mathrm{H})(\mathrm{X} \otimes \mathrm{Y})(\mathrm{Y} \otimes \mathrm{Z}) &= \mathrm{HXY} \otimes \mathrm{HYZ} \\
(\mathrm{H} \otimes \mathrm{H} \otimes \mathrm{X})(\mathrm{H} \otimes \mathrm{X} \otimes \mathrm{Y})(\mathrm{H} \otimes \mathrm{X} \otimes \mathrm{Z}) &= \mathrm{H}^3 \otimes \mathrm{HX}^2 \otimes \mathrm{XYZ}
\end{aligned}$$

- This property is useful for Pauli decomposition (decomposing a large matrix into Pauli axes: X, Y, Z) and computing the Hamiltonian matrix (taught in Chapter 7 afterwards)

- Step 0: The initial state is $|0\rangle^{\otimes N}$.

$$\left|s^{(0)}\right\rangle = |0\rangle \otimes |0\rangle \otimes |0\rangle$$

- Step 1: Apply the Hadamard gate on the first qubit.

$$\begin{aligned}\left|s^{(1)}\right\rangle &= (\mathbf{H}\,|0\rangle) \otimes |0\rangle \otimes |0\rangle \\ &= |+\rangle \otimes |0\rangle \otimes |0\rangle\end{aligned}$$

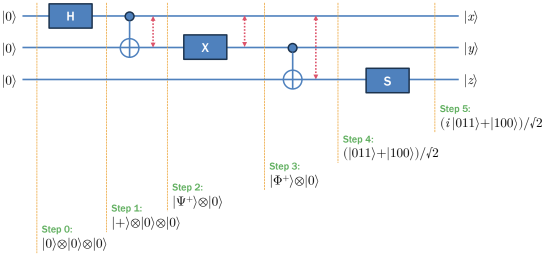


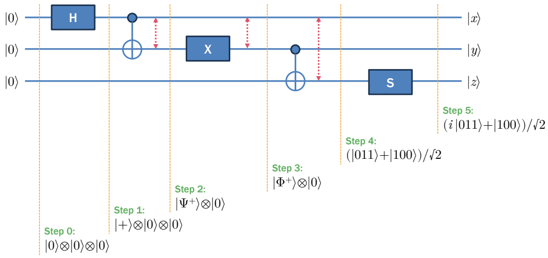
Figure 8: Another quantum algorithm

Quantum Algorithm: Example 2

- Step 2: Apply the CNOT gate on the first and second qubits.

$$\left|s^{(2)}\right\rangle = (\text{CNOT}\,(|+\rangle \otimes |0\rangle)) \otimes |0\rangle$$
$$= \left(\text{CNOT}\,\frac{|00\rangle + |10\rangle}{\sqrt{2}}\right) \otimes |0\rangle$$
$$= \frac{|00\rangle + |11\rangle}{\sqrt{2}} \otimes |0\rangle$$

The result is an entangled state, so a bidirectional arrow is shown.



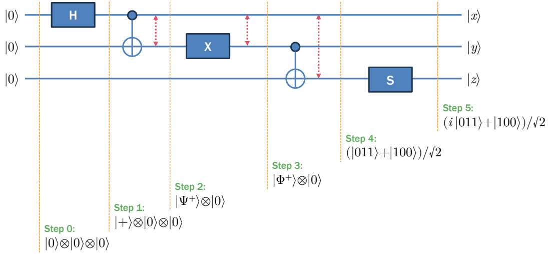
- Step 3: Apply the NOT gate on the second qubit. Since it is entangled with the first qubit, we have to augment the NOT gate.

$$\mathbf{I} \otimes \mathbf{X} \quad = \quad |x, \overline{y}\rangle \langle x, y|$$

Note that the first qubit is preserved while the second one is flipped.

$$\left|s^{(3)}\right\rangle \quad = \quad \left((\mathbf{I} \otimes \mathbf{X}) \frac{|00\rangle + |11\rangle}{\sqrt{2}} \right) \otimes |0\rangle$$

$$= \quad \frac{|01\rangle + |10\rangle}{\sqrt{2}} \otimes |0\rangle$$



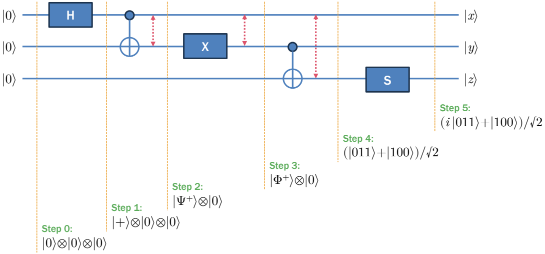
Quantum Algorithm: Example 2

- Step 4: Apply the CNOT gate on the second and third qubits. Since the second qubit is entangled with the first one, we augment the CNOT gate.

$$\mathrm{I} \otimes \mathrm{CNOT} \quad = \quad |x, y, y \oplus z\rangle \langle x, y, z|$$

Note that the first qubit is preserved.

$$\begin{aligned} \left| s^{(4)} \right\rangle &= (\mathrm{I} \otimes \mathrm{CNOT}) \left(\frac{|01\rangle + |10\rangle}{\sqrt{2}} \otimes |0\rangle \right) \\ &= (\mathrm{I} \otimes \mathrm{CNOT}) \frac{|010\rangle + |100\rangle}{\sqrt{2}} \\ &= \frac{|011\rangle + |100\rangle}{\sqrt{2}} \end{aligned}$$



Note that it results in another entangled state. That is why we mark it with a bidirectional arrow across the three qubits.

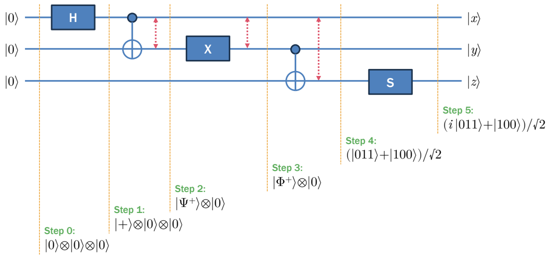
Quantum Algorithm: Example 2

- Step 5: Apply the phase shift gate on the third qubit. Since it is entangled with the first and second ones, we have to augment the gate.

$$
\begin{aligned}
\mathrm{I}^{\otimes 2} \otimes \mathrm{S} &= |x,y,0\rangle\langle x,y,0| \\
&\quad + i\,|x,y,1\rangle\langle x,y,1|
\end{aligned}
$$

  Note that the first and second qubits are preserved.

$$
\begin{aligned}
\left|s^{(5)}\right\rangle &= (\mathrm{I}^{\otimes 2} \otimes \mathrm{S}) \left(\frac{|011\rangle + |100\rangle}{\sqrt{2}} \right) \\
&= \frac{i\,|011\rangle + |100\rangle}{\sqrt{2}}
\end{aligned}
$$



Example 2 in PennyLane

```python
import pennylane as qml
from pennylane import numpy as np

dev = qml.device('default.qubit', wires=[0,1,2])

@qml.qnode(dev)
def circuit():
    # State preparation
    qml.H(wires=[0])
    qml.CNOT(wires=[0,1])
    qml.X(wires=[1])
    qml.CNOT(wires=[1,2])
    qml.S(wires=[2])

    # Measurement
    return qml.state()

print(circuit())
```

Result:

```
[0.    0.    0.    0.707j    0.707    0.    0.    0.]
```

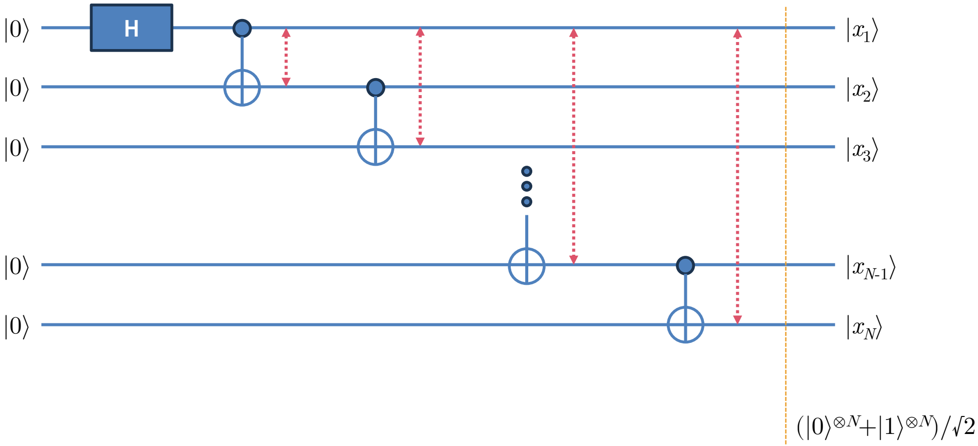


Figure 9: Hadamard gate and a series of CNOT can be used to create the GHZ state.

Exercise 3.2: Entanglement Preparation

Design quantum algorithms that prepare the following states. Develop PennyLane code to verify each of them.

1. $|\Psi^+\rangle$
2. $|\Phi^+\rangle$
3. $\frac{1}{\sqrt{2}}\left(|000\rangle + |111\rangle\right)$
4. $\frac{1}{\sqrt{2}}\left(|100\rangle + |011\rangle\right)$
5. $\frac{1}{\sqrt{2}}\left(|0011\rangle + |1100\rangle\right)$
6. $\frac{1}{\sqrt{2}}\left(|100\rangle - |011\rangle\right)$
7. $\frac{1}{\sqrt{2}}\left(|100\rangle - i\,|011\rangle\right)$
8. $\frac{1}{\sqrt{2}}\left(i\,|100\rangle - |011\rangle\right)$
9. $\frac{1}{\sqrt{2}}\left(i\,|101\rangle + |010\rangle\right) \otimes |\Psi^-\rangle$

Answer the following questions.

10. Let's define the adjoint shift gate

$$\mathbf{S}^* = |0\rangle\langle 0| - (i\,|1\rangle)\langle 1|$$

Show that $\mathbf{SSS}\,|x\rangle = \mathbf{S}^*\,|x\rangle$ for any quantum state $|x\rangle$.

11. Show that

$$\begin{aligned} \mathbf{I}^{\otimes M} \otimes \mathbf{U} &= \mathbf{I}_{2^M} \otimes \mathbf{U} \\ \mathbf{U} \otimes \mathbf{I}^{\otimes M} &= \mathbf{U} \otimes \mathbf{I}_{2^M} \end{aligned}$$

where $\mathbf{I}_M$ is the $M \times M$ identity matrix.

Clifford Set/2

Level 2: Pauli-X Gate

- Pauli gate rotates the qubit for phase π on one the axes: X, Y, and Z
- Pauli X flips the qubit

$$|0\rangle \mapsto |1\rangle \qquad |1\rangle \mapsto |0\rangle$$

which is equivalent to rotation on Axis X

$$\mathbf{X} = |1\rangle\langle 0| + |0\rangle\langle 1| = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- $\mathbf{X}$ is self-inversible: $\mathbf{XX}|x\rangle = |x\rangle$
- `qml.X(wires=...)`

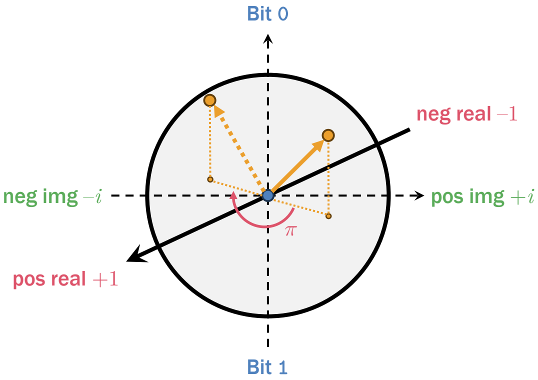


Figure 10: Pauli X gate

- Pauli Y flips the qubit, flips the sign, and flips the amplitude

$$|0\rangle \mapsto i\,|1\rangle \qquad |1\rangle \mapsto -i\,|0\rangle$$

which is equivalent to rotation on Axis Y

$$\mathbf{Y} = (i\,|1\rangle)\langle 0| + (-i\,|0\rangle)\langle 1| = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

- $\mathbf{Y}$ is self-inversible: $\mathbf{YY}\,|x\rangle = |x\rangle$
- `qml.Y(wires=…)`

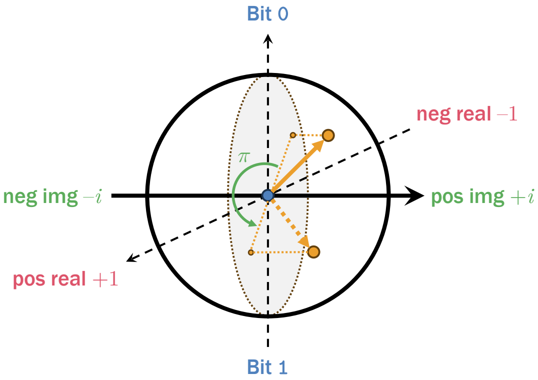


Figure 11: Pauli Y gate

Level 2: Pauli-Z Gate

- Pauli Z keeps the qubit and flips the sign

$$|0\rangle \mapsto |0\rangle \qquad |1\rangle \mapsto -|1\rangle$$

  while is equivalent to rotation on Axis Z

$$\mathbf{Z} = |0\rangle\langle 0| + (-|1\rangle)\langle 1| = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- $\mathbf{Z}$ is self-inversible: $\mathbf{ZZ}|x\rangle = |x\rangle$
- `qml.Z(wires=…)`

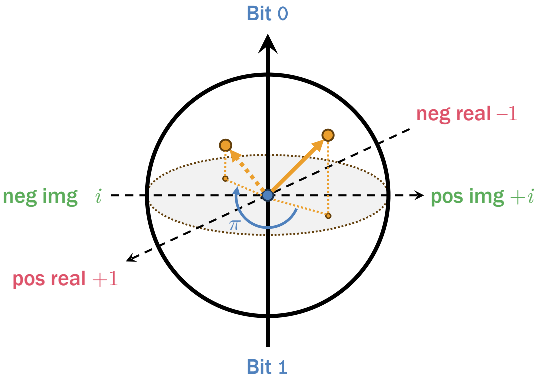


Figure 12: Pauli Z gate

Summary of Level 2

Gates	$\|0\rangle$	$\|1\rangle$	Flip qubit?	Flip sign?	Flip amplitude?
$\mathbf{X}$	$\|1\rangle$	$\|0\rangle$	Y	N	N
$\mathbf{Y}$	$i\,\|1\rangle$	$-i\,\|0\rangle$	Y	Y	Y
$\mathbf{Z}$	$\|0\rangle$	$-\|1\rangle$	N	Y	N

Table 4: Pauli gates. All of them belong to the $\mathbf{SU}(2)$ group.

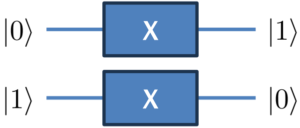


Figure 13: Pauli X

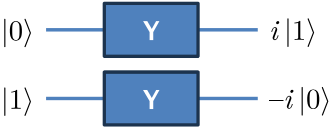


Figure 14: Pauli Y

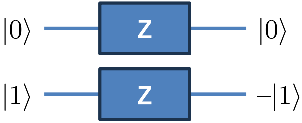


Figure 15: Pauli Z

Generalized Operator Augmentation

- For any quantum operator Q, we can augment it with trivial qubits by

$$\mathrm{Q}^{(N)}_{p_1,\dots,p_M} = \sum_{k=1}^{K} \left|x'_1,\dots,x'_N\right\rangle \left\langle x_1,\dots,x_N\right|$$

where the output mixed state is specified by the indices $p_1,\dots,p_M$, i.e.

$$\left|x'_{p_1},\dots,x'_{p_M}\right\rangle = \mathrm{Q}\left|x_{p_1},\dots,x_{p_M}\right\rangle$$

while the remaining input qubits are unchanged, i.e. $x'_{p_j} = x_{p_j}$ for all indices $j \notin \{p_1,\dots,p_M\}$

- Examples:

$$\begin{aligned}
\mathrm{X}^{(3)}_{3} &= |x,y,1\rangle\langle x,y,0| \\
&\quad + |x,y,0\rangle\langle x,y,1| \\
\mathrm{Z}^{(3)}_{2} &= |x,0,z\rangle\langle x,0,z| \\
&\quad + (-1)\,|x,1,z\rangle\langle x,1,z| \\
\mathrm{H}^{(3)}_{1} &= |+,y,z\rangle\langle 0,y,z| \\
&\quad + |-,y,z\rangle\langle 1,y,z| \\
\mathrm{CNOT}^{(4)}_{1,3} &= |0,x,y,z\rangle\langle 0,x,y,z| \\
&\quad + |1,x,\overline{y},z\rangle\langle 1,x,y,z|
\end{aligned}$$

Quantum Algorithm: Example 3

- Step 0: The initial state is $|0\rangle^{\otimes N}$.

$$\left|s^{(0)}\right\rangle = |0\rangle \otimes |0\rangle \otimes |0\rangle \otimes |0\rangle$$

- Step 1: We recognize two entangled states caused by the Hadamard and CNOT gates.

$$\left|s^{(1)}\right\rangle = \frac{1}{2}\left(|00\rangle + |11\rangle\right) \otimes \left(|00\rangle + |11\rangle\right)$$

- Step 2: We apply the X and Z gates on the first and latter pairs of qubits.

$$\begin{aligned}\left|s^{(2)}\right\rangle &= \left(\mathbf{I} \otimes \mathbf{X} \otimes \mathbf{I} \otimes \mathbf{Z}\right)\left|s^{(1)}\right\rangle \\ &= \frac{1}{2}\left(|01\rangle + |10\rangle\right) \otimes \left(|00\rangle - |11\rangle\right)\end{aligned}$$

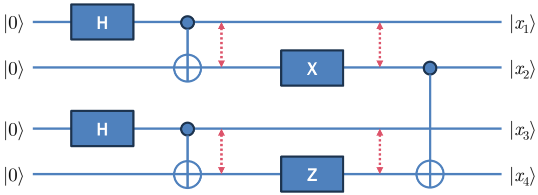


Figure 16: Yet another quantum algorithm

Quantum Algorithm: Example 3

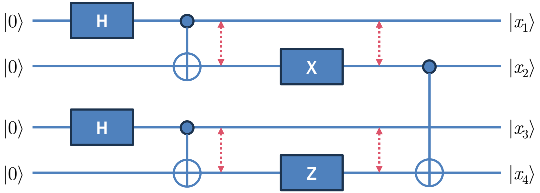
- Step 2 (cont'd): We expand the tensor product of $\left|s^{(2)}\right\rangle$

$$\begin{aligned}\left|s^{(2)}\right\rangle &= \frac{1}{2}(|0100\rangle - |0111\rangle \\ &\quad + |1000\rangle - |1011\rangle)\end{aligned}$$

- Step 3: The CNOT gate for the 2nd and 4th qubits is simply a CNOT gate with the 1st and 3th qubits augmented to it.

$$\mathrm{CNOT}^{(4)}_{2,4} = |v,x,y,x\oplus z\rangle\langle v,x,y,z|$$

Observe that v and y are introduced as dummy variables.



- Step 3 (cont'd): Applying this gate on $\left|s^{(2)}\right\rangle$ yields

$$\begin{aligned}\mathrm{CNOT}^{(4)}_{2,4}\left|s^{(2)}\right\rangle &= \frac{1}{2}(|0101\rangle - |0110\rangle \\ &\quad + |1000\rangle - |1011\rangle)\end{aligned}$$

which is no longer entangled.

Exercise 3.3: Pauli Gates

Design quantum algorithms that prepare the following states. Develop PennyLane code to verify each of them.

1. $|\Phi^+\rangle$
2. $\frac{i}{\sqrt{2}}(|01\rangle - |10\rangle)$
3. $|\Psi^-\rangle$
4. $\frac{i}{\sqrt{2}}(-|00\rangle + |11\rangle)$ [Use $\mathbf{Y}$.]
5. $\frac{1}{\sqrt{2}}(|011\rangle + |100\rangle)$ [Use $\mathbf{Z}$.]
6. $\frac{1}{\sqrt{2}}(-|010\rangle - i\,|101\rangle)$ [Use $\mathbf{Y}$.]
7. $\frac{i}{\sqrt{2}}(|0100\rangle - |1011\rangle)$ [Use $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$.]
8. $\frac{i}{2}(|0101\rangle - |0110\rangle - |1000\rangle + |1011\rangle)$ [Use $\mathbf{Y}$ and $\mathbf{Z}$.]

Answer the following questions.

9. Show that $\mathbf{Z}\,|x\rangle = \mathbf{HXH}\,|x\rangle$. It means that $\mathbf{Z}$ can be emulated by the Hadamard gate and the NOT gate.
10. Show that $\mathbf{Y}\,|x\rangle = i\mathbf{XZ}\,|x\rangle$. It means that $\mathbf{Y}$ can be emulated by the Hadamard gate and the NOT gate.
11. Show that $-i\mathbf{XYZ} = \mathbf{I}$.
12. Show that for any single-qubit quantum operator $\mathbf{Q}$,

$$\mathbf{Q}_j^{(N)} = \mathbf{I}^{\otimes j-1} \otimes \mathbf{Q} \otimes \mathbf{I}^{\otimes N-j}$$

Clifford Set/3

Level 3: SWAP Gate

- Swap: We can trivially swap places of two qubits for ease of processing

$$\begin{aligned}\mathrm{SWAP} &= |y,x\rangle\langle x,y| \\ &= |00\rangle\langle 00| + |10\rangle\langle 01| \\ &\quad + |01\rangle\langle 10| + |11\rangle\langle 11| \\ &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\end{aligned}$$

- This is useful for rearranging the qubits before passing them to a quantum gate
- `qml.SWAP(wires=…)`

- Example:

$$\mathrm{SWAP}\left(\frac{|01\rangle - |10\rangle}{\sqrt{2}}\right) = \frac{|10\rangle - |01\rangle}{\sqrt{2}}$$

- SWAP is self-inversible

$$\mathrm{SWAP}\,\mathrm{SWAP}\,|x,y\rangle = |x,y\rangle$$

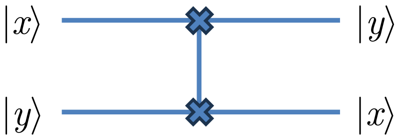


Figure 17: Swap gate

Level 3: Controlled Z Gate

- Controlled Z gate keeps the qubit and flips the sign if the control qubit is $|1\rangle$

$$
\begin{aligned}
\mathbf{CZ} &= (-1)^{x \cdot y} |xy\rangle \langle xy| \\
&= |00\rangle\langle 00| + |01\rangle\langle 01| \\
&\quad + |10\rangle\langle 10| + (-1)|11\rangle\langle 11| \\
&= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}
\end{aligned}
$$

- $\mathbf{CZ}$ is self-inversible: $\mathbf{CZ}\,\mathbf{CZ}|x,y\rangle = |x,y\rangle$

- `qml.CZ(wires=…)`
- Example:

$$\mathbf{CZ}\left(\frac{|01\rangle - |10\rangle}{\sqrt{2}}\right) = \frac{|01\rangle - |11\rangle}{\sqrt{2}}$$

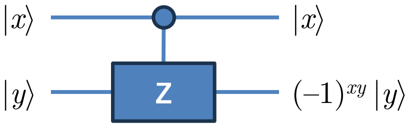


Figure 18: Controlled Z gate

Exercise 3.4: Swap and Controlled Z

Answer the following questions.

1. Let's define the Controlled Y gate $\mathbf{CY}$ from the following state mapping.

   $$|0x\rangle \mapsto |0x\rangle \quad |10\rangle \mapsto i\,|10\rangle \quad |11\rangle \mapsto -i\,|11\rangle$$

   Explain why $\mathbf{CY}$ is not self-inversible.

2. Compute $\mathbf{SWAP}^{(4)}_{2,3}\,|s\rangle$, where

   $$|s\rangle \quad = \quad \frac{1}{2}\left(|00\rangle + |11\rangle\right) \otimes \left(|01\rangle + |10\rangle\right)$$

3. Create $\frac{1}{\sqrt{2}}(|0\rangle^{\otimes N} + (-1)^{N+1}\,|1\rangle^{\otimes N})$, where $N \geq 1$, using only $\mathbf{H}$ and $\mathbf{CZ}$.

4. Show that $\mathbf{CZ} = (\mathbf{I} \otimes \mathbf{H})\ \mathbf{CNOT}\ (\mathbf{I} \otimes \mathbf{H})$. It means that $\mathbf{CZ}$ can be emulated by the $\mathbf{H}$ and $\mathbf{CNOT}$ gates.

5. Show that $\mathbf{SWAP} = \mathbf{CNOT}^{(2)}_{1,2}\ \mathbf{CNOT}^{(2)}_{2,1}\ \mathbf{CNOT}^{(2)}_{1,2}$. It means that $\mathbf{SWAP}$ can be emulated only by the $\mathbf{CNOT}$ gate.

6. Design a quantum operator $\mathbf{L}$ that rotates a mixed state of N qubits to the left for one bit. E.g. $\mathbf{L}\,|101100\rangle = |011001\rangle$. Then convert $\mathbf{L}$ to a quantum algorithm.

Circuit Visualization

Circuit Visualization

- We can visualize a quantum circuit in two formats: text and graphics
- Text format: `qml.draw(circuit)(params)`
- This instruction displays the given circuit after applying the given parameters on it

```python
@qml.qnode(dev)
def circuit(theta: float):
    qml.H(wires=[0])
    qml.RY(theta, wires=[0])
    qml.CNOT(wires=[0,1])
    return qml.state()

print(qml.draw(circuit)(np.pi / 6))
```

```
0: ──H──RY(0.52)─╭●─┤  State
1: ──────────────╰X─┤  State
```

Figure 19: String representation of a quantum circuit

Circuit Visualization

- Graphical format: `qml.draw_mpl(circuit)(params)`

```
@qml.qnode(dev)
    def circuit(theta: float):
    qml.H(wires=[0])
    qml.RY(theta, wires=[0])
    qml.CNOT(wires=[0,1])
    return qml.state()

fig, ax = qml.draw_mpl(circuit)(np.pi / 6)
fig.savefig('circuit.png')
```

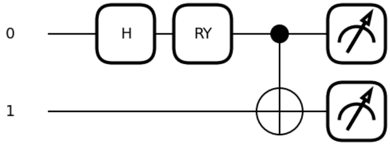


Figure 20: Graphical representation of a quantum circuit

Conclusion

Clifford Set of Quantum Gates

Gates	Symbols	From $\|0\rangle$	From $\|1\rangle$
NOT	$\mathbf{X}$	$\|1\rangle$	$\|0\rangle$
Hadamard	$\mathbf{H}$	$\frac{\|0\rangle+\|1\rangle}{\sqrt{2}}$	$\frac{\|0\rangle-\|1\rangle}{\sqrt{2}}$
Phase shift	$\mathbf{S}$	$\|0\rangle$	$i\|1\rangle$
Pauli-X	$\mathbf{X}$	$\|1\rangle$	$\|0\rangle$
Pauli-Y	$\mathbf{Y}$	$i\|1\rangle$	$-i\|0\rangle$
Pauli-Z	$\mathbf{Z}$	$\|0\rangle$	$-\|1\rangle$

Table 5: Single-qubit gates

Gates	Symbols	Descriptions
CNOT	$\mathbf{CNOT}$	$\|0,x\rangle \mapsto \|0,x\rangle$ $\|1,x\rangle \mapsto \|1,\overline{x}\rangle$
Swap	$\mathbf{SWAP}$	$\|x,y\rangle \mapsto \|y,x\rangle$
CZ	$\mathbf{CZ}$	$\|0,x\rangle \mapsto \|0,x\rangle$ $\|1,x\rangle \mapsto \|1,\mathbf{Z}\,x\rangle$

Figure 21: Two-qubit gates

Operator Augmentation

- If we have a quantum operator

$$\mathbf{Q} = \sum_{k=1}^{K} |\phi_k\rangle \langle\psi_k|$$

we can augment $\mathbf{Q}$ with trivial qubits by

$$\mathbf{I}^M \otimes \mathbf{Q} = \sum_{k=1}^{K} |x_1, \dots, x_M, \phi_k\rangle \langle x_1, \dots, x_M, \psi_k|$$

$$\mathbf{Q} \otimes \mathbf{I}^M = \sum_{k=1}^{K} |\phi_k, x_1, \dots, x_M\rangle \langle\psi_k, x_1, \dots, x_M|$$

- This is useful for accommodating all entangled qubits, some of which not involved with the operator

- For any quantum operator $\mathbf{Q}$, we can augment it with trivial qubits by

$$\mathbf{Q}^{(N)}_{p_1,\dots,p_M} = \sum_{k=1}^{K} \left|x'_1, \dots, x'_N\right\rangle \langle x_1, \dots, x_N|$$

where the output mixed state is specified by the indices $p_1, \dots, p_M$, i.e.

$$\left|x'_{p_1}, \dots, x'_{p_M}\right\rangle = \mathbf{Q}\left|x_{p_1}, \dots, x_{p_M}\right\rangle$$

while the remaining input qubits are unchanged, i.e. $x'_{p_j} = x_{p_j}$ for all indices $j \notin \{p_1, \dots, p_M\}$

Questions?

Target Learning Outcomes/1

- Conceptual Understanding of Quantum Circuits
  1. Define the three primary stages of a quantum algorithm: state preparation, computation through gate manipulation, and measurement.
  2. Explain the characteristics of the Clifford gate set, including fault-tolerance, self-inversibility, and efficient classical simulability.
- Quantum Gate Mastery
  1. Apply Level 1 and 2 gates (Hadamard, Phase Shift, and Pauli X,Y,Z) to transform pure states into superpositions or modify quantum phases.
  2. Differentiate between the effects of Pauli X,Y, and Z gates in terms of qubit flipping, sign flipping, and amplitude changes.
  3. Utilize multi-qubit gates (CNOT, SWAP, CZ) to create entanglement or rearrange qubit order for processing.

Target Learning Outcomes/2

- Mathematical and Algebraic Application
  1. <u>Calculate</u> the state of a quantum system after a series of gate applications using tensor products and matrix multiplication.
  2. <u>Perform</u> operator augmentation to describe the action of a gate on specific indices within an N-qubit state.
  3. <u>Prove</u> gate equivalencies through algebraic manipulation.
- Computational Implementation
  1. <u>Construct</u> functional quantum circuits using the PennyLane library, including device initialization and QNode definition.
  2. <u>Visualize</u> quantum circuits in both text and graphical formats to verify algorithm structure.
  3. <u>Develop</u> algorithms to prepare specific quantum states, such as Bell states or GHZ states.